

Unmanned Ground Vehicle (UGV) - Defense Bot

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Abstract— Military troops operating in hostile and unfriendly territories are exposed to the risk of being mortally wounded by tactile explosives such as exploding mines, IEDCs, unexplored ordnance, etc. Even after wars, land mines are still a residual threat which may potentially wound an innocent traveler, civilian, etc., when the said individual is traversing the terrain, deprived of awareness with respect to the threat lying ahead. To avoid such situations, the problem under investigation can be addressed by employing unmanned ground vehicles. Unmanned vehicles [1] which are monitored, operated and controlled by military personnel, who need not be present in the terrain of activity, is a platform for multiple payloads such as mine detection and clearance, supply delivery systems and also have the option to accommodate mountable weapons such machine guns with acoustic suppression while firing. Defense bot is mounted with a special type of gun called as an electro-magnetic gun which works on the principle of electro-magnetic attraction towards Ferro-magnetic substances, using which Ferro-magnetic projectiles can be fired. Defense bot is capable of blending into the enemy territory, owing to the credibility of the robot's compactness. As an additional feature, camouflaging techniques can be implemented to further enhance its stealth capabilities.

Keywords- Defense Bot, UGV, Military, Robotic System, Ammunition

I. INTRODUCTION

Military robots, drones, unmanned vehicles, etc. are used for fighting enemy forces and organizations, intelligence gathering such as taking visual images of enemy infrastructures and strongholds, surveillance and reconnaissance and rescue missions [2]

Customized drones according to requirements can be employed to fulfill necessities in the aspects of military defense and offense applications [3]. These remotely operable robotic systems are immune to the threats posed by environments which, on a contrary basis are actually threatening to Human personnel, whose intervention in the said environments may be of utmost importance. To avoid such issues, the UGV systems prove to be of great help regarding their capability of being deployable to such hostile environments.

Here we have a custom-built UGV [4], dubbed as 'Defense bot' with respect to the project title. This robotic system contains a day-and-night vision camera which proves to be useful in surveillance and reconnaissance activities. A key feature proving to be of use here is a stealth gun which fires Ferro-magnetic projectiles and is mounted in a turret which is operated with a customized remote control [5]

For the purpose of aiming, we have a laser dot targeting system. Here we can operate the bot within a terrain radius of 100m from the operator remote control. The operable defense range of the bot can be enhanced by using Very High Frequency (VHF) communication system or satellite communication.

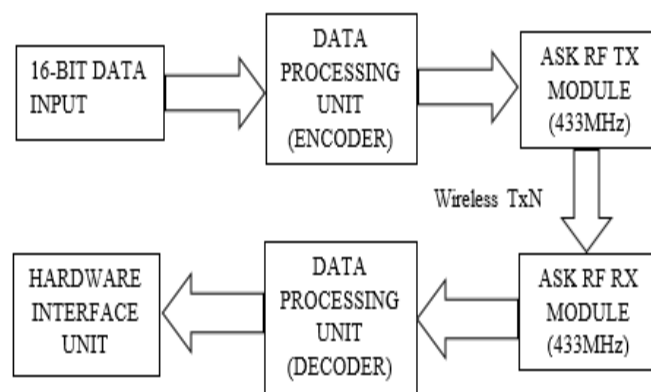


Fig 1. General Block Diagram

Figure 1 describes defense bot's [6] general functioning block diagram. We need a 16-bit communication channel for the operation and control of defense bot. The processing unit for data on the transmitter end, comprises of encoders and parallel-to-serial converters which are suitable for modulating the carrier wave. Data processing unit is core of wireless control. The processing of data results in the assignment of the desired address to 16-bit input signals provided. The technique of modulation employed here Amplitude Shift Keying (ASK) and the modulation signal frequency is 433MHz. The range of modulation can be increased by using VHF modulation for long range operating requirements. The low-pass (LPF) filter circuit present in the RF demodulator circuit performs partial conversion of analog signals to digital signals and the data-processing unit present at the receiver side, will acquire these signals. Referring to the receiver setup, the unit which processes the received data, consists of a converter which gives a parallel data output after converting serial data which was given to it as an input, followed by a decoder. The signals upon decoding will be able to drive the other hardware present in the system. Motor-driver IC, solid-state switch, bi-stable transistor switch, etc. are some of the circuit elements contained within the control unit.

II. TRANSMITTER CIRCUITRY

The components under transmitter includes the following,

- A. HD12E Encoder
- B. ASK RF Transmitter Module

The in-depth transmitter block is shown under Fig 2

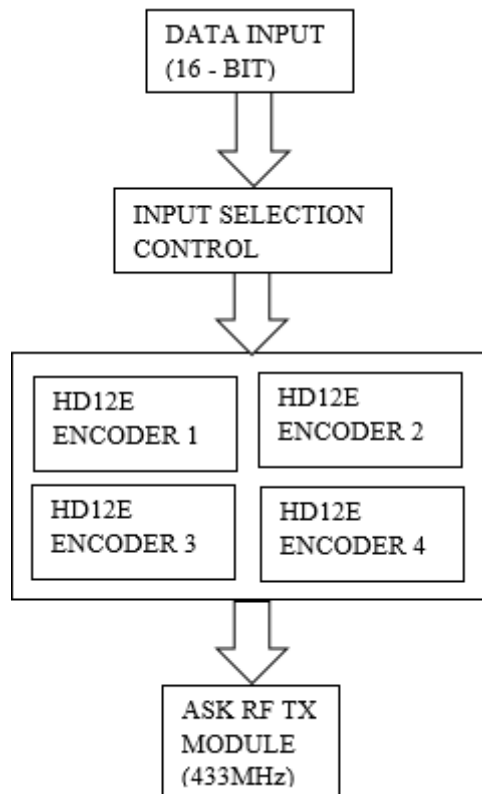


Fig 2. In-detail Transmitter Block Diagram

A. HD12E ENCODER

The HD12E IC is a 12-bit encoder. Its main use is well founded in Tx-Rx setups where the IC is paired with its decoding counterpart i.e., with a 12-bit decoder. HD12E operating process of the input data involves the conversion of parallel inputs to serial output. For carrying out successful transmission of data through an RF transmitter, it is required that the input parallel data of 12 bits be converted to serial data. The 12 bits of interest here, are segmented into an address bit field of 8 bits and a data bit field of 4 bits. With respect to this project, the so mentioned IC will convert 4-bit parallel data to 4-bit serial data with respect to a designated address of choice.

B. AMPLITUDE SHIFT KEYING RADIO-FREQUENCY TRANSMITTER MODULE

ASK RF module responsible for generating a sine wave carrier of a high frequency. When a digital high is given as input, the transmitter module will transmit high frequency carrier signal with suitable amplitude and similarly, when a digital low is recognized at the input of the module, there is no carrier wave transmitted for the duration of the provided input signal.

III. RECEIVER CIRCUITRY

The components under receiver includes the following,

- A. HD12E Decoder
- B. ASK RF Receiver Module
- C. Capacitor Bank
- D. Motor Driver Unit
- E. DC to DC Converter
- F. Solid State Switch (SCR)
- G. Charge and Trigger Control Unit
- H. Turret Control Unit
- I. Lithium – Polymer (LiPo) Battery
- J. Single pole double throw double pole double throw (SPDTPDPT) relay
- K. Power MOSFETs (IRFZ44)
- L. Push-pull configuration fly-back transformer

The in-depth receiver block is shown under Fig 3

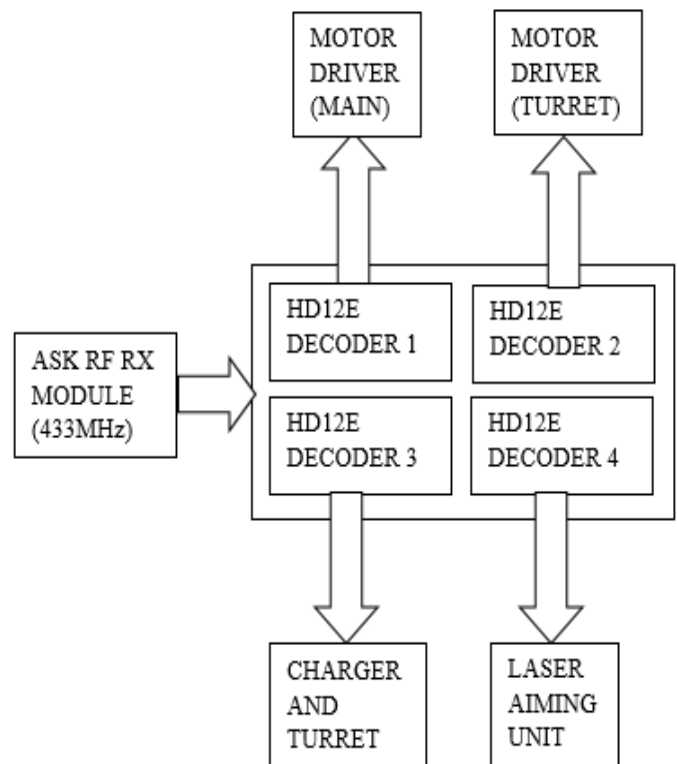


Fig 3. In-detail Receiver Block Diagram

A. HD12E DECODER

HD12D is a 12-bit decoder [7]. The IC comprises of 18 pins. The 18 pins are assigned with the following designations: 8 address pins, a pin for serial data input, 4 pins that are required for parallel data output, VDD, VSS, osc1 and osc2 pins which are used for an inbuilt oscillator and for setting a decoder frequency, a valid transmission pin which is used to monitor the completion and validation of transmission. The internal hardware of the IC is designed in a way where the data will be received at the data out pins, provided the address codes received match with the decoder local address assigned to it. Defense Bot contains 4 such HD12D ICs. The data output of each of these ICs are tethered to the motor driver module, turret control module, charge and trigger control module and the laser aiming module through circuit connections.

B. AMPLITUDE SHIFT KEYING RADIO-FREQUENCY RECEIVER MODULE

ASK RF receiver module demodulates the received ASK signals and converts them to digital signals [8]. The frequency of the carrier used here is 433MHz. The demodulated signal is acquired by the DATA IN of all four HD12D decoder ICs and the address codes in the demodulated signal is checked and the data is outputted by the decoder which has a similar local address as that of the received address. The ASK RF Receiver module comprises of a low-pass filter circuit which is used to convert sine wave of high frequency into digital pulse signal. The digital pulses are further provided to HD12D and motor driver.

C. CAPACITOR BANK

A Capacitor Bank consists of multiple capacitors which are connected in series, parallel or series-parallel combination to store electrical energy [9]. The bank is then put to use so as to maintain a power factor or correct the phase shift in a DC power source. They are also exploited to be used in a DC power source, where the amount of total energy stored can be increased. With respect to our project, we require a quantifiable capacitance of 2000 μ F.

D. MOTOR DRIVER UNIT

This is a dual inline package IC. It contains two H-bridge circuits embedded inside the IC. The dual H-bridge circuits fulfill the act of driving the two motors in two different directions. The IC of interest needs 4-bit data input to drive two motors in different directions [10]. This IC possesses a capacity to deliver 600mA of current continuously and 1.2amps of peak current. In this project, the use of this IC is to drive two pairs of gear motors. Also, we are driving a turret controlled motor by employing this IC.

E. DC-TO- DC CONVERTER

The DC-to-DC converter is basically ZVS (zero voltage switching) resonant power converter which has a very high efficiency which reaches an appreciable range of 80 to 90%, with very less switching losses [11]. The charger circuit consists of capacitor voltage monitoring circuit which is basically an op-amp Schmidt trigger. The instant when the desired voltage level becomes equal to the voltage present across the capacitor, the op-amp Schmidt trigger will shut down the charging process.

F. SOLID STATE SWITCH (SCR)

70TPS12 is an SCR (silicon controlled rectifier or thyristor) which is designed with the desire of exploiting high and medium power switching, phase control applications, etc. Other applications involve input rectification, AC-switches, phase-control operations, etc. Threshold voltage V_T at 100 A is < 1.4 V. Maximum repetitive peak voltage, V_{RRM} is 1200/1600 V [12]. Here, in this project, the accelerating coil is triggered by using the said component. This SCR basically fulfills the duty of a solid state switch, which will conduct several kilo-amps for a small temporal span.

G. CHARGE AND TRIGGER CONTROL UNIT

The sudden discharge of the capacitor bank creates a surge current which causes the bullet to be accelerated from the electro-magnetic gun due to electro-magnetic attraction. The capacitor bank's routine of charge and discharge is controlled by the third decoder [13].

H. TURRET CONTROL UNIT

Turret of the defense bot consists of a camera which is suitable to provide vision in the day as well as in the night, laser aiming assist and electro-magnetic gun. The turret is driven by high torque low R.P.M gear motors. The turret in the bot has a 360° freedom of movement in the horizontal plane and 45° freedom of movement in the vertical plane.

I. LITHIUM POLYMER (LIPO) BATTERY

Lithium Polymer (Li-Po) battery is a DC power source having a cell voltage of 4.2V. This battery possesses a high power density, as well as high energy density. Adding to the battery's characteristic specification, it has long charge-discharge cycle life [15].

The battery we are using with respect to the context of the project, has a charge-burst rate of 25C. Power rating of the battery is 2.2Ah.

J. SINGLE POLE DOUBLE THROW DOUBLE POLE DOUBLE THROW (SPDTDPDT) RELAY

A relay can be interpreted as an electro-mechanical switch. It consists of an electro-magnet inside which will actuate the contactor movement to open and close the circuit. In this project, the discussed relay is fully put to use in turning on and off the capacitor charger circuit based on the received signals from the op-amp Schmidt trigger. It controls the gate signal for the power MOSFETs used in the project.

K. POWER MOSFETs (IRFZ44)

Power MOSFETs can handle few amperes to hundreds of amperes. In this case, the power MOSFET under consideration is capable of handling fifty amperes of current under room temperature [14]. So, this particular power MOSFET is employed to drive a push-pull configuration transformer, at high frequencies of operation.

L. PUSH-PULL CONFIGURATION FLY-BACK TRANSFORMER

This particular transformer is ideal for high power applications. In the case of this project it gives two hundred watts of power at the output. The operation frequency of this particular transformer is twenty kilohertz to thirty kilohertz. The advantage of using this transformer is that for a small sized core we can get a significantly high output power. The discussed transformer is used to charge a capacitor bank till 450V.

IV. METHODOLOGY

The electro-magnetic gun in the defense bot works based on attraction towards Ferro-magnetic substance. The Ferro-magnetic projectile which is to be fired will be located at the breach of the electro-magnetic accelerating coil. The capacitor takes the responsibility of generating a high current surge, which is directed through the electro-magnet. The resulting magnetic field will influence the projectile in a way that it reaches the center of the coil. This short burst of high acceleration achieved between the breach and the coil's center will give enough momentum to the projectile to continue the linear motion. The capacitor bank which gives the required energy to the coil, is charged by a DC-to-DC step-up converter which converts 12V DC to 450V DC. The electro-magnetic coil exploits the electro-static energy stored in the capacitor by converting it into mechanical energy.

The remote control of the defense bot consists of many input sections such as robot motion control, turret motion control, charge and discharge control and laser aiming. The projectile's speed was found to be 22m/s through mathematical calculations and deductions performed upon the data obtained from the optical sensors. This has been observed by using optical sensors which are kept a meter apart. When the bullet is released, it produces sine waves with a time delay and that time delay is used in calculating the velocity of the bullet. Here the mass of the bullet is approx. 10g.

Mass of projectile = 10g

Velocity of projectile = 40 m/s

Total inductance, $L = 30\mu\text{H}$

Total kinetic energy ($\text{KE}_{\text{total}} = \frac{1}{2} \times 0.010 \times (40)^2 = 8\text{J}$)

Total input energy = $\frac{1}{2} \times 2000 \times 10^{-6} \times (400)^2 = 160\text{J}$

Total capacitance = $2000\mu\text{F}$

Total efficiency for single stage = $\frac{8}{160} = 5\%$

Total current flowing through the coil,

$$I = \frac{V}{R} (1 - e^{-Rt/L})$$

$$I = \frac{400}{0.2} \times (1 - e^{-0.2 \times 400 \times 10^{-6} / 30 \times 10^{-6}})$$

$$I = 1861.0\text{A}$$

$$\text{RC time constant, } t = 0.2 \times 2000 \times 10^{-6}$$

$$t = 400 \times 10^{-6}$$

V. VISUAL DEMONSTRATION

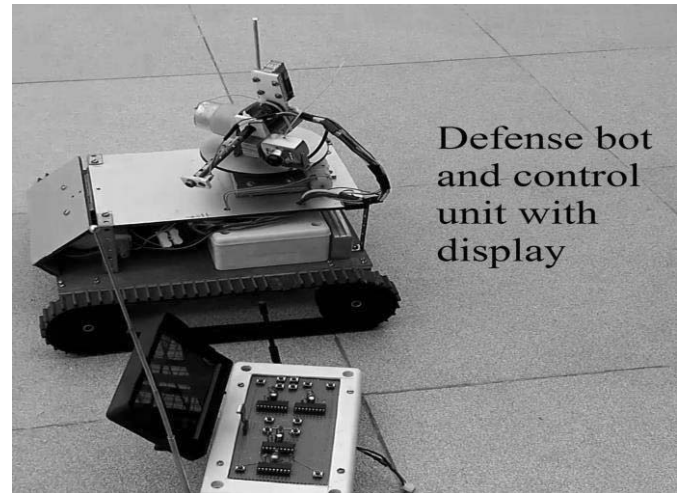


Fig 4a. Defense bot and control unit set – up

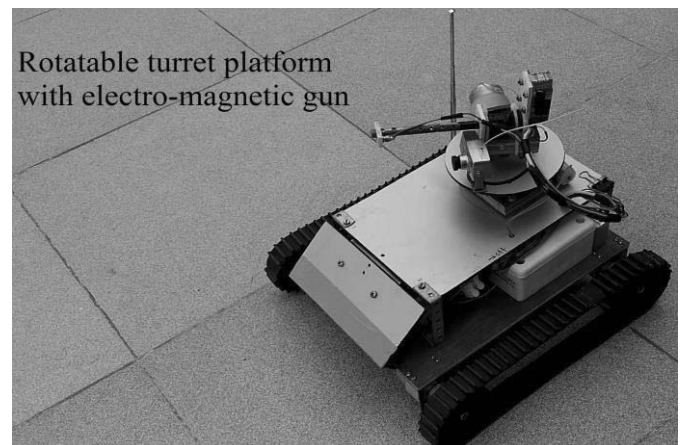


Fig 4b. Rotatable head mount electromagnetic gun

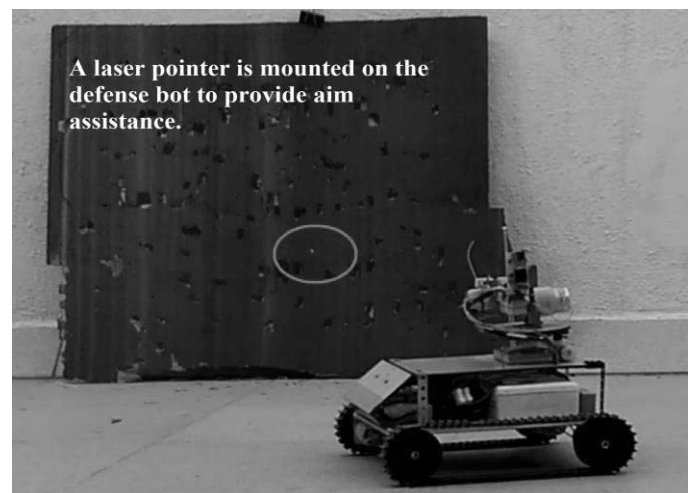


Fig 4c. Laser pointer locked to the target by the heat mount electromagnetic gun

VI. CURRENT APPLICATIONS

The current prototype can be used for the following applications,

- i. Border Patrolling and enemy territory surveillance
- ii. Operation in Bio-hazard area and radioactive hazard area
- iii. Operations involving spying, search and destroy

VII. FUTURE SCOPE

The robot can be improvised to traverse vertical surfaces using amplified Van der Waals force of attraction principle. The mobility on different vertical surfaces can be achieved using claw – gripping effect to climb on porous surfaces such as brick and concrete walls, mud walls etc. and electromagnetic surface attraction to climb on metallic walls. The mobility can be improved on air, land and water for stealth operations.

The maximum effective range of the present gun is 10mts and we can extend it up to a range of 50mts. In addition to this, different types of lethal and non-lethal ammunitions such as armor-piercing rounds, anti-material rounds, Taser rounds etc can be implemented on the defense bot.

The current firing rate is limited to 20 projectiles per minute. It can be increased by using high energy density batteries, which has very high discharge rate in the range of 100-200C.

The defense bot can be modified to possess automated all – terrain mobility based on GPS and obstacle sensors. The defense bot module can be appended to an IoT system for controlling from a remotely localized place.

VIII. CONCLUSION

The electromagnetic gun designed in the defense bot performed as per the design specifications and the efficiency was calculated to be 5%. The efficiency can be increased by using multiple stage of acceleration for the given input energy. The mobility of the defense bot covers all aspects of traversing various terrains of the horizontal plane for a given altitude level of a tangible platform meant for movements.

The rate of firing of the electromagnetic gun mounted on the defense bot proved to be sufficient to demonstrate the stealth – offensive capabilities on observation. The current rate of firing is 20 projectiles per minute. After which, the module requires manual re-loading of ammunition.

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